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BOARD EXAM REPEATED & HIGH-PRIORITY QUESTION BANK

Chapter: Molecular Basis of Inheritance • Categorized 1M, 2M, 3M & 5M Model Answers & Value Points

SECTION A: 5-MARK LONG ANSWER QUESTIONS (HIGH PROBABILITY)

Target 5/5 with Labelled Schematic Proofs

Q1. Describe the salient features of Watson and Crick's Double Helix Model of B-DNA.

5 MARKS

- **Two Polynucleotide Chains:** Backbones made of alternating pentose sugars and phosphate units; nitrogenous bases project inwardly towards the helix axis.
- **Antiparallel Polarity:** One chain runs in the 5' → 3' orientation while the complementary chain runs in the 3' → 5' direction.
- **Complementary Base Pairing:** Purines pair strictly with pyrimidines via hydrogen bonds: **A = T** (2 H-bonds) and **G ≡ C** (3 H-bonds).
- **Helical Dimensions:** Right-handed helix with a pitch of **3.4 nm (34 Å)**, containing **10 base pairs per turn** (axial rise = **0.34 nm / 3.4 Å** per bp). Uniform helix diameter = **2.0 nm (20 Å)**.
- **Double Stability:** Conferred by hydrogen bonding between bases and hydrophobic plane-stacking interactions of aromatic bases over each other.

Board Diagram Checklist: Draw helical strands with 5'→3' and 3'→5' labels, show 20 Å diameter, 3.4 Å rise, major & minor grooves.

Q2. Explain the Lac Operon model in E. coli in both the absence and presence of lactose (inducer).

5 MARKS

- **Constitutive Regulator (i gene):** Synthesizes active *Lac Repressor protein* continuously.
- **Absence of Inducer (Switched OFF):** Active repressor binds firmly to the *operator (o)* region → physically blocks RNA Polymerase from transcribing structural genes (z, y, a) → No enzymes produced.
- **Presence of Inducer (Switched ON):** Lactose (or allolactose) binds to the repressor protein, inducing a conformational change that inactivates it → Inactive repressor cannot bind operator → RNA Polymerase binds promoter and transcribes polycistronic mRNA.
- **Enzymes Produced:**
  - **z gene:** β-galactosidase (hydrolyzes lactose → glucose + galactose).
  - **y gene:** β-galactoside permease (increases cell membrane permeability to lactose).
  - **a gene:** β-galactoside transacetylase.

Board Diagram Checklist: Draw [p-i-p-o-z-y-a] cassette and show Repressor-Operator binding (OFF) vs Inducer-Repressor complex (ON).

Q3. Describe the Hershey and Chase blender experiment that proved DNA is the genetic material.

5 MARKS

- **Objective:** To determine whether viral protein or DNA enters bacteria during bacteriophage infection.
- **Experimental Setup:**
  - Bacteriophages cultured with ^{35}S to label *viral coat proteins* (sulfur absent in DNA).
  - Bacteriophages cultured with ^{32}P to label *viral DNA core* (phosphorus absent in proteins).
- **Three Consecutive Steps:**
  1. **Infection:** Labeled phages were allowed to attach and infect *E. coli* bacteria.
  2. **Blending:** Agitated in a high-speed blender to strip off empty viral capsids from bacterial surfaces.
  3. **Centrifugation:** Spun down to separate heavy bacterial cells (pellet) from light viral coats (supernatant).
- **Observations & Conclusion:** Bacteria infected with ^{32}P showed radioactive pellets, while ^{35}S remained exclusively in the supernatant. Proved unequivocally that **DNA enters the host cell and acts as the genetic material**.

Q4. Explain the steps of DNA Fingerprinting technique with its applications. 5 MARKS

- **Principle:** Exploits high-degree allelic polymorphism in non-coding satellite DNA (**VNTRs - Variable Number of Tandem Repeats**, 0.1–20 kb). Developed by *Alec Jeffreys*.
- **Methodological Steps:**
  1. *DNA Extraction:* Isolation of genomic DNA from blood, semen, hair follicle, or tissue.
  2. *Restriction Digestion:* Cleavage using specific restriction endonucleases.
  3. *Electrophoresis:* Separation of DNA fragments based on molecular size in agarose gel.
  4. *Southern Blotting:* Denaturation and capillary transfer of DNA fragments onto a nylon or nitrocellulose membrane.
  5. *Hybridization:* Incubating the membrane with radiolabeled VNTR probe.
  6. *Autoradiography:* Exposure to X-ray film to generate individual-specific banding patterns.
- **Applications:** Forensic crime investigation, resolving disputed paternity/maternity, and tracing evolutionary biodiversity.

SECTION B: 3-MARK SHORT ANSWER QUESTIONS (CORE CONCEPTS)

Point-to-Point Value Marks

Q5. Describe the structure of a eukaryotic nucleosome core with a neat labelled diagram. 3 MARKS

- **Histone Octamer Core:** Composed of 2 units each of four basic proteins (H2A, H2B, H3, H4) rich in positively charged amino acids **Lysine and Arginine**.
- **DNA Wrapping:** Negatively charged DNA (~200 bp) wraps 1.65 turns around the positively charged octamer.
- **H1 Linker Histone:** Placed at the entry and exit points to stabilize the repeating "beads-on-a-string" nucleosome unit.

Q6. Differentiate between Euchromatin and Heterochromatin. 3 MARKS

Feature	Euchromatin	Heterochromatin
Packing & Staining	Loosely coiled, light-staining region	Densely coiled, dark-staining region
Transcriptional Status	**Transcriptionally active**	**Transcriptionally inactive (silent)**
Replication Stage	Replicates early during S phase	Replicates late during S phase

Q7. Describe the post-transcriptional modifications occurring in eukaryotic hnRNA. 3 MARKS

- **Splicing:** Non-coding sequences (*introns*) are cleaved out and coding sequences (*exons*) are joined covalently by spliceosomes.
- **5' Capping:** An unusual nucleotide, **7-methylguanosine triphosphate (m^7Gppp)**, is attached to the 5'-end.
- **3' Tailing (Polyadenylation):** 200–300 adenylate residues are added to the 3'-end in a template-independent manner to generate mature, export-ready mRNA.

Q8. List any six salient features of the Genetic Code. 3 MARKS

- **Triplet nature:** 3 adjacent bases constitute a codon (61 sense codons + 3 stop codons).
- **Unambiguous & Specific:** One particular codon specifies only one single amino acid.
- **Degenerate:** Some amino acids are coded by more than one codon.
- **Universal:** Same codon specifies identical amino acid across all living taxa (e.g., UUU = Phenylalanine from bacteria to humans).
- **Commaless:** Read continuously 5' → 3' without any intervening punctuation.
- **Dual function of AUG:** Initiator start codon and encodes **Methionine (Met)**.

Q9. How did Meselson and Stahl prove that DNA replication is semi-conservative? 3 MARKS

- Grew *E. coli* in $^{15}NH_4Cl$ medium to label parental DNA heavy (^{15}N-^{15}N).
- Transferred cells into normal ^{14}N medium and extracted DNA across generations using **CsCl density gradient centrifugation**.
- **Results:** Generation I (20 min) yielded **100% hybrid DNA (^{15}N-^{14}N)**; Generation II (40 min) yielded **50% light (^{14}N-^{14}N) and 50% hybrid (^{15}N-^{14}N)**, validating semi-conservative transmission.

SECTION C: 2-MARK & 1-MARK RAPID-FIRE BOARD QUESTIONS

Definitive 1-2 Line Statements

Q10. Why is DNA considered chemically more stable than RNA? **2 MARKS**

1. DNA lacks a reactive 2'-OH group on its deoxyribose sugar, making it chemically inert compared to labile RNA.
2. Presence of **Thymine (5-methyluracil)** in place of uracil confers added resistance to oxidative/thermal degradation.

Q11. State Chargaff's rules of base equivalence. **2 MARKS**

For any double-stranded DNA: (1) Total Purines = Total Pyrimidines ($[A]+[G] = [T]+[C]$), (2) Stoichiometric ratio $[A]/[T] = 1$ and $[G]/[C] = 1$. Base ratio $([A]+[T])/([G]+[C])$ is constant for a given species.

Q12. What are UTRs? State their significance. **2 MARKS**

Untranslated Regions (UTRs) are mRNA sequences flanking the coding region (at 5' end before start codon and at 3' end after stop codon). They are essential for **efficient translation initiation and mRNA stability**.

Q13. Rapid 1-Marker Board Fillers: **1 MARK EACH**

- **Name the stop / nonsense codons:** UAA (ochre), UAG (amber), UGA (opal).
- **Name the enzyme acting as ribozyme in bacteria:** 23S rRNA (catalyzes peptidyl transfer).
- **Largest human gene:** *Dystrophin* on X-chromosome (2.4 million base pairs).
- **What is Teminism?** Reverse transcription (synthesis of DNA from RNA template via reverse transcriptase).
- **Bacterial initiation & termination factors:** σ (sigma) factor for initiation; ρ (rho) factor for termination.
- **Haploid DNA content in humans:** $3.3 \times 10^9\text{ bp}$ ($6.6 \times 10^9\text{ bp}$ in diploid cells).